

(i) Estimation

Students should be taught to:

estimate population means from samples;

estimate population proportions from samples
with applications in opinion polls and
elsewhere;

**estimate population size based on the
Petersen capture/recapture method;**

understand the effect of sample size on
estimates and the variability of estimates,

**with a simple quantitative appreciation of
appropriate sample size.**

**The appropriateness of the assumptions
in practice.**

3 Reasoning, interpreting and discussing results

Content

Students should be taught in the context of real data to:

apply statistical reasoning, explain and justify inferences, deductions, arguments, solutions **and decisions**;

explore connections, look for and examine relationships between variables **including fitting the equation to a line of best fit or trend line**;

consider the limitations of any assumptions;

formally identify outliers using quartiles;

relate summarised data to any initial questions or observations;

interpret all forms of statistical tables, diagrams and graphs;

compare distributions of data and make comparisons using measures of central tendency and measures of dispersion, percentiles, **deciles, inter-percentile range, mean deviation, variance and standard deviation**;

check results for reasonableness and modify their approaches if necessary;

interpret correlation as a measure of the strength of the association between two variables, **including Spearman's rank correlation coefficient for ranked data**;

make predictions;

compare or choose by eye between a line of best fit and a model based on $y = ax^n + b$ for $n = 2, 1$ or $\frac{1}{2}$, $y = ax^2 + bx$ or $y = ka^x$.

Notes

Cases clearly restricted to the content of the specification at the appropriate level

Eg height and weight, age and depreciation of a car, GNP and mortality in infants.

Interpretations of gradient and intercept will be expected.

Simple cases only, eg honest replies to questionnaires, equally likely outcomes in probabilities, representativeness of sample of population, reliability of secondary data.

Dealing with outliers will be expected.

The relevance of measures of central tendency.

To include real published tables and graphs.

The shapes of distributions and graphs may be used. Formula for variance and standard deviation to be given.

Eg the mean must lie between the maximum and minimum, 'the average bicycle speed was 130 km per hour' is not reasonable.

The use of words such as weak or strong will be expected; **the closer to ± 1 the better the correlation for a given sample size. Beware the use of correlation in small samples.**

The use of a trend line by eye, drawing or formula will be expected.

Based on an informal awareness of the spread of points around a proposed model.

4 Probability

Content

Students should be taught to:

understand the meaning of the words event and outcome;

understand words such as impossible, certain, highly likely, likely, unlikely, possible, evens and present these on a likelihood scale;

put outcomes in order in terms of probability;

put probabilities in order on a probability scale;

understand the terms ‘random’ and ‘equally likely’;

understand and use measures of probability from a theoretical perspective and from a limiting frequency or experimental approach;

the relationship between ‘odds’ and probability;

understand that in some cases the measure of probability based on limiting frequency is the only viable measure;

compare expected frequencies and actual frequencies;

use simple cases of the binomial and discrete uniform distribution;

use simulation to estimate more complex probabilities;

use probability to assess risk;

produce, understand and use a sample space;

understand and use Venn diagrams and cartesian grids;

Notes

Tossing a coin is an event with outcomes landing Heads or Tails.

Interpretation of real-life situations will be expected, eg ‘the probability that the horse will win the next race is 0.3’; ‘the probability that I will get a grade C or better in my Statistics is $\frac{3}{4}$.’

Use of $\leq$ will be expected.

Labelling of the scale will be expected.

Formal definition and notation of a limit will not be required whilst terminology such as ‘as the number of trials increases’ will be required. **Understand that increasing sample size generally leads to better estimates of probability and population parameters.**

The relationship between $\Sigma p = 1$ and ‘odds’ may be tested.

The probability of a sports team winning can only be measured from a limiting frequency perspective. For example, medical statistics for assessment of health risks.

The expansion of $(p + q)^2$ will be expected. In all other cases the expansion of $(p + q)^n$ will be given.

Examples may be taken from insurance.

Listing all outcomes of single events and two successive events, in a systematic way.

Content

understand the terms mutually exclusive and exhaustive and to understand the addition law
 $P(A \text{ or } B) = P(A) + P(B)$;

know, for mutually exclusive outcomes, that the sum of the probabilities is one and in particular the probability of something not happening is one minus the probability of it happening;

form and use tree diagrams and probability tree diagrams for independent events **and conditional cases**;

understand, use and apply the addition, **general addition** and multiplication laws for independent events **and conditional events and outcomes**;

the shape and simple properties of the normal distribution.

Notes

$$P(A \text{ or } B) = P(A) + P(B);$$

‘Mutually exclusive’ means that the occurrence of one outcome prevents another, $\Sigma = 1$ when summed over all mutually exclusive outcomes.

$$\text{If } P(A) = p \text{ then } P(\text{not } A) = 1 - p.$$

Listing all possible joint or compound outcomes **with and without replacement for up to three outcomes and three sets of branches.**

To correctly apply

$$P(A \text{ and } B) = P(A) \times P(B),$$

$$P(A \text{ or } B) = P(A) + P(B)$$

$$P(A \cup B) = P(A) \times P(B) - P(A \cap B),$$

$$P(A \text{ or } B) = P(B | A) \times P(A).$$

The distribution is symmetrical with mean, mode and median equal; approximately 95% of values are within ± 2 standard deviations of the mean; virtually all values are within ± 3 standard deviations of the mean. Use of the normal distribution to model some populations.

Use of Normal distribution tables will not be required.

Edexcel GCSE Statistics

Formulae Sheet

Foundation Tier

$$\text{Mean of a frequency distribution} = \frac{\sum fx}{\sum f}.$$

$$\text{Mean of a grouped frequency distribution} = \frac{\sum fx}{\sum f}, \text{ where } x \text{ is the mid-interval value.}$$

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Formulae Sheet

Higher Tier

Mean of a frequency distribution $= \frac{\sum fx}{\sum f}$.

Mean of a grouped frequency distribution $= \frac{\sum fx}{\sum f}$, where x is the mid-interval value.

Variance $= \frac{\sum (x - \bar{x})^2}{n}$

Standard deviation (set of numbers) $\sqrt{\left[\frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2 \right]}$

or $\sqrt{\left[\frac{\sum (x - \bar{x})^2}{n} \right]}$
where $\bar{x}$ is the mean set of values.

Standard deviation (discrete frequency distribution) $\sqrt{\left[\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2 \right]}$

or $\sqrt{\left[\frac{\sum f(x - \bar{x})^2}{\sum f} \right]}$

Spearman's rank correlation coefficient $1 - \frac{6\sum d^2}{n(n^2 - 1)}$